

Day 7 Appointment Module

Date

August 27, 2026

Overview

Implemented the Appointment module for booking and managing patient-doctor appointments. Added appointment entity with status tracking, REST and view controllers, and role-based access for booking operations.

Tech Stack

- Java 26
- Spring Boot 4.1.0
- Lombok 1.18.46
- MongoDB (clinic_db)
- Thymeleaf + Bootstrap 5 + Thymeleaf-SpringSecurity6
- Spring Security with @PreAuthorize
- JWT for REST API + Session for Browser

Files Created

New Java Files

1. `com.clinic.enums.AppointmentStatus` - Enum: SCHEDULED, COMPLETED, CANCELLED
2. `com.clinic.entity.Appointment` - Appointment document with patientId, doctorId, date, time, status, notes
3. `com.clinic.dto.AppointmentRequestDto` - DTO for booking appointments with validation
4. `com.clinic.dto.AppointmentResponseDto` - DTO with patientName and doctorName resolved
5. `com.clinic.exception.AppointmentNotFoundException` - Custom exception
6. `com.clinic.repository.AppointmentRepository` - MongoDB repository with `findByPatientId`, `findByDoctorId`
7. `com.clinic.service.AppointmentService` - Interface for appointment operations
8. `com.clinic.service.AppointmentServiceImpl` - Implementation with patient/doctor name resolution
9. `com.clinic.controller.AppointmentRestController` - REST endpoints with role-based access
10. `com.clinic.controller.AppointmentViewController` - Thymeleaf pages for appointments

New HTML Files

1. `templates/appointments/list.html` - Appointment list with status badges and role-based actions
2. `templates/appointments/form.html` - Book/edit appointment form with patient/doctor dropdowns
3. `templates/appointments/detail.html` - Appointment detail view

Updated Files

1. `templates/index.html` - Added Appointments card and navbar link
2. `templates/patients/list.html` - Added Appointments to navbar
3. `templates/patients/detail.html` - Added Appointments to navbar

4. `templates/doctors/list.html` - Added Appointments to navbar
5. `templates/doctors/detail.html` - Added Appointments to navbar
6. `templates/auth/login.html` - Added Appointments to navbar
7. `templates/auth/register.html` - Added Appointments to navbar
8. `com.clinic.config.SecurityConfig` - Added `/appointments/**` to `permitAll`
9. `com.clinic.repository.PatientRepository` - Added `findByPatientId`, `findByNameContainingIgnoreCase` with `Pageable`
10. `com.clinic.repository.DoctorRepository` - Fixed `findByDoctorId` return type
11. `com.clinic.service.PatientServiceImpl` - Updated to use `findByPatientId`
12. `com.clinic.service.DoctorServiceImpl` - Added `@Service` annotation

REST API Endpoints

Method	Endpoint	Allowed Roles
GET	<code>/api/appointments</code>	ADMIN, DOCTOR, RECEPTIONIST
GET	<code>/api/appointments/{id}</code>	ADMIN, DOCTOR, RECEPTIONIST
GET	<code>/api/appointments/patient/{patientId}</code>	ADMIN, DOCTOR, RECEPTIONIST
GET	<code>/api/appointments/doctor/{doctorId}</code>	ADMIN, DOCTOR, RECEPTIONIST
POST	<code>/api/appointments</code>	ADMIN, RECEPTIONIST
PUT	<code>/api/appointments/{id}</code>	ADMIN, RECEPTIONIST
PATCH	<code>/api/appointments/{id}/status</code>	ADMIN, DOCTOR, RECEPTIONIST
DELETE	<code>/api/appointments/{id}</code>	ADMIN

Thymeleaf Pages

URL	Description	Role Access
<code>/appointments</code>	List all appointments	All roles
<code>/appointments/new</code>	Book new appointment	ADMIN, RECEPTIONIST
<code>/appointments/{id}</code>	View appointment details	All roles
<code>/appointments/{id}/edit</code>	Edit appointment	ADMIN, RECEPTIONIST
<code>/appointments/{id}/delete</code>	Delete appointment	ADMIN

Key Features Implemented

1. Appointment booking with patient and doctor dropdowns
2. Status tracking: SCHEDULED (yellow), COMPLETED (green), CANCELLED (red)
3. Appointment ID format: APT-XXXXXXXX
4. Patient and doctor names resolved in response DTOs
5. Status update via PATCH endpoint
6. Form validation on all required fields
7. Role-based button visibility on appointment pages
8. Status color-coded badges in Bootstrap

Testing

Browser Tests

- Book appointment with patient, doctor, date, time, notes
- View appointment details
- Edit appointment (change time/status)
- Status badge colors: yellow=SCHEDULED, green=COMPLETED, red=CANCELLED
- Role-based hiding: ADMIN sees all, DOCTOR views only, RECEPTIONIST books/edits

API Tests

- POST /api/appointments with Bearer token → 201 Created
- PATCH /api/appointments/{id}/status?status=COMPLETED → Updates status
- GET /api/appointments/patient/{patientId} → Filtered by patient
- GET /api/appointments/doctor/{doctorId} → Filtered by doctor

Validation Tests

- Empty form submission → Red error messages on all required fields

Issues Faced & Solutions

Issue 1: DoctorRepository findByDoctorId wrong return type

- Error: ScopedValue instead of Optional
- Solution: Fixed return type to Optional

Issue 2: PatientRepository missing Pageable overload

- Error: findByNameContainingIgnoreCase(String, Pageable) not found
- Solution: Added Page findByNameContainingIgnoreCase(String, Pageable)

Issue 3: DoctorService bean not found

- Error: AppointmentViewController required DoctorService bean
- Solution: Added @Service annotation to DoctorServiceImpl

Git Commit

feat: add appointment module with booking, status tracking, and role-based access